MILITARY SPECIFICATION
CARTRIDGE CASE AND BASE PLUG
40MM, XM199E2

1. SCOPE

1.1 This specification covers Cartridge Case and Base Plug, 40MM, XM199E2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for

STANDARDS

Federal

Federal Test Method Standard No. 151 - Metals; Test Methods

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-1167	- Ammunition Data Card
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

Munitions Command

C10542607	- Case, Cartridge 40MM, XM199E2
C10542393	- Plug, Base
IEL-8648436	- Index of Inspection Equipment Lists for Cartridge Case and Base Plug 40MM, XM199E2

FSC 1310

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

ASA-B46.1 - Surface texture

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N.Y.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications, and the following:

3.2 Extrusion rod soundness.- The extrusion rods supplied for the manufacture of cartridge cases shall be free from cracks, including cracks existing in the ingot or billet and partially welded during extrusion, splits, inclusions and gross porosity.

3.3 Metal defects.- All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any other defects.

3.4 Mechanical properties.- The mechanical properties specified on the applicable drawings and referenced specifications shall be uniform throughout each component.

3.5 Protective coating.- Protective coating shall be in accordance with the requirements of the applicable drawing and specifications listed thereon.

3.6 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.6.1 Burr.- No part shall have a burr which might interfere with the assembly or function of the round, or which might be injurious to personnel handling the item.

3.6.2 Foreign matter.- No part shall contain dirt, chips, grease, oil, rust, corrosion or other foreign matter.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define Quality Assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of:

15 cartridge cases (coated)
15 cartridge cases (uncoated)
15 base plugs (coated)
15 base plugs (uncoated)

The sample shall be manufactured in the same manner using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all the requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.- Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

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4.3 Inspection.

4.3.1 Lot.

4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.3.1.2 Solution heat treatment lot.- Solution heat treatment lots shall consist of one heat treated furnace load or fraction thereof of a batch type furnace. If a continuous type of solution heat treat furnace is used, a lot shall consist of a maximum of 5000 cartridge cases or fraction thereof of continuous production. The lots thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects containing herein, using the sampling plans contained in MIL-STD-105. Continuous sampling plans in accordance with standard MIL-STD-1235 may be used if approved by the procuring activity. An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be used. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination. The roughness comparison specimens prescribed by ASA-B46.1 shall be used as a basis for surface roughness determination.

4.3.2.1 Tool control dimensions.- Dimensions marked tool control shall be gaged at the beginning of each eight hours production from each machine in operation and prior to any tooling change. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted, provided the contractor has established correlation between the tool dimension and the component dimensions prior to the start of production. The contractor shall be responsible for maintaining adequate surveillance to assure that drawing requirements are being met throughout the life of the tool. All material produced prior to periodic inspection will be rejected and screened whenever a defective tool control dimension is found.

4.3.2.2 Case, Cartridge, Drawing C10542607.

Defect	Method of Inspection
CRITICAL	
1. Propellant flash hole(s) missing or obstructed	Visual

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4.3.2.2 (Cont'd)

	Defect	Method of Inspection
MAJOR		
101.	Diameter of sabot cavity not functionally concentric with outside body diameter	Gage
102.	Diameter of sabot cavity counterbore not functionally concentric withoutside body diameter	Gage
103.	Diameter of flange not functionally concentric with outside body diameter	Gage
104.	Diameter of base plug cavity counterbore not functionally concentric with outside body diameter	Gage
105.	Diameter of base plug cavity not functionally concentric with base plug cavity counterbore	Gage
106.	Rear face not perpendicular to outside body diameter	Gage
107.	Diameter of propellant flash holes	Gage
108.	Diameter of body (outside)	Gage(See 6.3)
109.	Diameter of sabot cavity counterbore	Gage
110.	Diameter of sabot cavity	Gage
111.	Diameter of flange	Gage
112.	Diameter of base plug cavity counterbore	Gage
113.	Diameter of base plug cavity	Gage
114.	Depth of sabot cavity counterbore	Gage
115.	Depth from rear face to bottom of base plug cavity	Gage
116.	Length from rear face to bottom of sabot cavity	Gage
117.	Thickness of flange	Gage
118.	Metal defective	Visual
MINOR		
201.	Depth to base plug cavity counterbore	Gage
202.	Length, total	Gage
203.	Location of propellant flash holes	Gage
204.	Depth of undercut forward of flange (if applicable)	Visual (See 6.1)
205.	Width of undercut forward of flange (if applicable)	Visual (See 6.1)
206.	Radii or chamfers missing or incorrect	Visual (See 6.1)
207.	Surface finish improper	Visual (See 6.1)
208.	Protective coating inadequate or damaged	Visual
209.	Color improper	Visual
210.	Marking missing, incorrect or illegible	Visual
211.	Burr	Visual
212.	Foreign matter	Visual

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4.3.2.3 Plug, Base, Drawing C10542393.

Defect	Method of Inspection
CRITICAL	
1. Primer flash hole missing or obstructed	Visual
MAJOR	
101. Concentricity of forward outside diameter with flange	Gage
102. Concentricity of primer cavity with flange	Gage
103. Perpendicularity of rear face with flange diameter	Gage
104. Diameter of propellant cavity	Gage
105. Diameter at front (outside)	Gage
106. Diameter of primer cavity	Gage
107. Diameter of primer flash hole	Gage
108. Diameter of flange	Gage
109. Depth of primer cavity	Gage
110. Length from rear face to datum diameter	Gage
111. Length from rear face to front of flange	Gage
112. Thickness through base	Gage
113. Metal defective	Visual
MINOR	
201. Diameter of propellant cavity counterbore	Gage
202. Depth of propellant cavity counterbore	Gage
203. Length, total	Gage
204. Width of flange	Gage
205. Radii or chamfers missing or incorrect	Visual (See 6.1)
206. Surface finish improper	Visual (See 6.1)
207. Protective coating inadequate or damaged	Visual
208. Burr	Visual
209. Foreign matter	Visual

4.3.3 Maximum diameter gage.- Each cartridge case shall be gaged in a maximum diameter free fall chamber gage. Failure of the cartridge case to comply with the requirements of the applicable drawing shall be cause for rejection of the cartridge case. (See 6.3)

4.3.4 Inspection testing.- The following tests shall be performed in accordance with 4.4.

4.3.4.1 Extrusion rod soundness.- Each rod extruded from each billet (ingot) shall be subjected to a macro etch test to determine compliance with 3.2. Failure of any macro etched specimen to comply shall be cause for rejection of the rod represented by the specimen.

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4.3.4.2 Mechanical properties.

4.3.4.2.1 Hardness.

4.3.4.2.1.1 Solution heat treatment lot.- Twenty five cartridge cases from each solution heat treatment lot (as defined by 4.3.1.2), shall be selected at random and subjected to a Rockwell Hardness test or by any other method approved by the procuring activity, to assess uniformity of the furnace. Hardness values shall be recorded.

4.3.4.2.2 Tension test.

4.3.4.2.2.1 Cartridge case.- Two cartridge cases, one of the hardest and one of the softest as determined by the hardness test shall be subjected to this test. Failure of any tension test specimen to comply with the mechanical properties requirements of the applicable drawing shall be cause for rejection of the lot represented by the sample. Solution heat treated lots rejected for not meeting minimum mechanical properties requirements may, at the discretion of the contractor, be reheat treated and hardness tests and tension tests outlined in 4.3.4.2.1.1 and 4.3.4.2.2.1 be submitted for acceptance.

4.3.4.2.2.2 Base plug.- The selection of samples for the tension test of material for the base plug shall be as specified in the applicable material specification. Failure of any tension test specimen to comply with the mechanical properties requirement shall be cause for rejection of the material represented by the sample.

4.3.4.3 Protective coating.- A sample size of five base plugs and five cartridge cases per each lot shall be tested for compliance to the requirements of the applicable specification. Failure to comply with the requirements shall be cause for rejection of the lot represented by the sample.

4.3.5 Inspection equipment.- Index of Inspection Equipment List No. IEL-8648436 identifies the inspection equipment required to perform the examination and tests prescribed herein. Equipment design shall be in accordance with code designation on the applicable IEL. (Reference is made to Inspection Equipment List for definition of coding format.)

4.4 Test methods and procedures.

4.4.1 Extrusion rod soundness.- A specimen shall be removed from each end of each rod, after extrusion from the ingot or billet. The specimens, after suitable surface preparation, shall be macro etched in a ten percent solution of sodium hydroxide and water heated to a temperature of 160 degrees F for a minimum of ten minutes after which the specimen shall be visually examined to determine compliance with the requirement.

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4.4.2 Mechanical properties.

4.4.2.1 Hardness test.-From the samples supplied in accordance with 4.3.4.2.1.1, the hardness test will be performed on the rear face of the cartridge cases. The test shall be performed using the equipment and procedures prescribed by Federal Test Method Standard No. 151.

4.4.2.2 Tension test.

4.4.2.2.1 Cartridge case.-From the samples supplied in accordance with 4.3.4.2.2.1, one tension test specimen conforming to the largest obtainable standard "F" type specimen as prescribed by Federal Test Method Standard No. 151 shall be taken in a circumferentially transverse direction from the side wall of the cartridge case forward of the web area. The number of specimens and their location are for the convenience of the contractor and Government and shall not restrict the contractor or Government from testing additional specimens at other locations to determine that the mechanical properties of the cartridge case meet the drawing requirement. Procedures and criteria for determining mechanical properties shall be as prescribed by Federal Test Method Standard No. 151.

4.4.2.2.2 Base plug.-From the samples supplied in accordance with 4.3.4.2.2.2, one longitudinal tension test specimen conforming to the largest obtainable standard "R" type specimen as prescribed by Federal Test Method Standard No. 151 shall be taken and tested. If base plugs are machined from annealed rod, the rod section selected for the tension test shall be heat treated with the represented base plugs after which one longitudinal tension test specimen conforming to the largest obtainable standard "R" type specimen as prescribed by Federal Test Method Standard No. 151 shall be taken and tested. The number of specimens and their location are for the convenience of the contractor and Government and shall not restrict the contractor or Government from testing additional specimens at other locations to determine that the mechanical properties requirements are met. Procedures and criteria for determining mechanical properties shall be as prescribed by Federal Test Method Standard No. 151.

4.4.3 Protective coating.-The procedures for testing the cleaning of components and the application of protective coating shall be in accordance with the criteria for testing as specified in applicable specification.

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5. PREPARATION FOR DELIVERY

5.1 Packing and marking.

5.1.1 Level A.-Not applicable.

5.1.2 Level B.-Not applicable.

5.1.3 Level C.-Packing and marking shall be in accordance with MIL-STD-1169.

5.2 Data card.-Data card information shall be MIL-STD-1167.

6. NOTES

6.1 Visual examination qualification.-When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.2 Ordering data.-Procurement documents should specify the title, number and date of this specification.

6.3 If the requirement specified in 4.3.3 cannot be met in free state, the "go" inside diameter plug gage may be inserted in the cartridge case while inspecting the "go" and "not go" outside body diameter. If the "go" inside plug is utilized, the data cards shall be annotated accordingly.

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Preparing activity

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